

Project Design Phase-I

Solution Architecture

Date	13 June 2026	Team ID	N/A (Solo Project)
Project Name	Network Anomaly Detection	Maximum Marks	4 Marks

Solution Architecture:

Solution Architecture Diagram — Network Anomaly Detection Data Flow



Figure 1: Architecture and data flow of the Network Anomaly Detection system using Isolation Forest on the CICIDS 2017 dataset

Key Architectural Decisions & Specifications

Component	Technology / Choice	Specification / Reason
Dataset	CICIDS 2017 (CSV)	Industry-standard benchmark; 80+ features; contains normal and attack traffic
Data Cleaning	Pandas	Remove null values, duplicate rows, and irrelevant columns
Feature Encoding	LabelEncoder (sklearn)	Convert categorical text columns (e.g. protocol type) to numeric values
Feature Scaling	StandardScaler (sklearn)	Normalise all features to equal ranges; prevents dominance by high-range features
Detection Algorithm	Isolation Forest (sklearn)	Unsupervised; no labelled attack data needed; fast; interpretable anomaly scores
No. of Trees	n_estimators = 100	Standard default; balances accuracy and computational speed
Anomaly Threshold	contamination = 0.05	Assumes ~5% of traffic is anomalous; tunable based on dataset analysis
Visualization	Matplotlib / Plotly	Lightweight plotting; no server infrastructure needed for academic MVP
Development Phase	Single-phase MVP	Solo project; all components built and tested in one development cycle